



NCCA

An Chomhairle Náisiúnta
Curraíom agus Measúnachta
National Council for
Curriculum and Assessment

Guidelines to support the Engineering Design and Manufacture Project

Leaving Certificate
Engineering

December 2025

Contents

INTRODUCTION	1
ASSESSMENT FOR CERTIFICATION IN LEAVING CERTIFICATE ENGINEERING	1
OVERVIEW OF THE DESIGN AND MANUFACTURE PROJECT	1
PROCESS FOR THE COMPLETION OF THE DESIGN AND MANUFACTURE PROJECT	2
Stage 1: Analysis of Brief, Brainstorming and Developing a Work Plan	3
Stage 2: Identify the focus of enquiry/research to be pursued	4
Stage 3: Development of an Optimum Solution	5
Stage 4: Manufacture of Physical Model of Optimum Solution.....	6
Stage 5: Testing, Evaluating and Refining	7
Stage 6: Finalising Physical Model and Folio	8
FORMAT AND SUBMISSION OF THE DESIGN AND MANUFACTURE PROJECT	9
Descriptors of Quality	10
ROLE OF THE TEACHER	11
APPENDIX 1: GUIDELINES TO SUPPORT REFERENCING.....	13
APPENDIX 2: SUGGESTED PROMPT QUESTIONS TO SUPPORT STUDENTS IN THE PROCESS FOR COMPLETION OF THE DESIGN AND MANUFACTURE PROJECT	15
Stage 1: Analysis of Brief, Brainstorming and Developing a Work Plan.....	15
Stage 2: Identify the focus of enquiry/research to be pursued	15
Stage 3: Development of an Optimum Solution.....	16
Stage 4: Manufacture of Physical Model of Optimum Solution.....	16
Stage 5: Testing, Evaluating and Refining.....	17
Stage 6: Finalising Physical Model and Folio.....	17
APPENDIX 3: PROTOTYPES	18

Introduction

This document, *Guidelines for the Leaving Certificate Engineering Design and Manufacture Project* provides:

- details of the nature and scope of the Design and Manufacture Project as outlined in the curriculum specification for Engineering
- guidance for schools, teachers and students on undertaking, completing and submitting the Design and Manufacture Project
- information on the role of schools and teachers in supporting students with the Design and Manufacture Project
- descriptors of quality for the Design and Manufacture Project.

These guidelines should be used in conjunction with the curriculum specification for Leaving Certificate Engineering which can be accessed at <https://www.curriculumonline.ie/senior-cycle/senior-cycle-subjects/engineering/>.

A brief for the conduct of the Design and Manufacture Project will be published annually by the State Examinations Commission (SEC) in term 1 of year 2.

Assessment for Certification in Leaving Certificate Engineering

Assessment for certification is based on the rationale, aim and learning outcomes of the Leaving Certificate Engineering specification. There are two assessment components: a written examination and an additional assessment component comprising of a Design and Manufacture Project. The written examination will be at higher and ordinary level. The Design and Manufacture Project will be based on a common brief and will be assessed at the level at which the student sits the final written examination. Each component will be set and examined by the SEC.

Table 1: Overview of assessment for certification

Assessment component	Weighting	Level
Design and manufacture project	50%	Common Brief
Written examination	50%	Higher and Ordinary Level

Overview of the Design and Manufacture Project

The Design and Manufacture Project provides an opportunity for students to display evidence of their learning across all strands of the specification. The senior cycle key competencies, developed through working with learning outcomes across the specification, will be applied through the student's engagement with the project. It has been designed to be naturally integrated into the teaching and learning of Engineering, making it a motivating and relevant experience for students. It also aims to draw together learning outcomes and cross-cutting

themes of the specification, while emphasising the importance of adopting and developing an Engineering mindset.

The Design and Manufacture Project brief will be issued annually by the SEC. The common brief will set out the requirements for the Design and Manufacture Project and will:

- set a context for the project
- provide guidance to students in the development of their project work
- allow students to develop their knowledge and understanding in areas related to the brief
- facilitate teachers and students in their planning.

Through their engagement with the Design and Manufacture Project, students will be able to demonstrate their creativity, showcase the breadth and depth of their practical and manufacturing ability, and refine their communication techniques as they develop, implement, and document their progress through the design and manufacturing process.

The Design and Manufacture Project will involve the following:

- analysing the brief and conducting independent research. Students will engage with real world sources, critically evaluate information, and use it to formulate an engineering solution and project plan
- developing ideas and making informed decisions. This includes selecting materials and components, designing and justifying an engineered solution, and creating a manufacturing plan
- manufacturing a unique, optimum solution. Students will apply the full breadth of their learning, carefully selecting materials and processes to produce a fully functioning model that demonstrates a range and depth of manufacturing skills and techniques that they have developed during engagement with the specification
- testing and evaluating the completed solution against the initial brief
- documenting the entire process in their design folio.

Upon completion, students will submit a solution and a supporting folio documenting the design and manufacturing process through an individual submission in a format specified by the SEC. The AAC will be marked by the SEC.

Schools have a high degree of autonomy in planning and organising the completion of the Design and Manufacture Project. This document, *Guidelines to support the Engineering Design and Manufacture Project*, gives guidance on a range of matters related to the organisation, implementation, and oversight of the project.

Process for the completion of the Design and Manufacture Project

The Design and Manufacture Project is designed to naturally integrate into everyday teaching and learning in classrooms. Through their study of Leaving Certificate Engineering, students have frequent opportunities to develop knowledge, skills, values and dispositions that are applicable to

the completion of the Design and Manufacture Project. Students are supported in thinking and behaving like engineers, fostering an engineering mindset that blends creativity, technical understanding, and reflective practice.

Teachers play a key role in helping students connect their day-to-day classroom activities to the demands of the project. This ongoing engagement builds a strong foundation of practical skills and theoretical knowledge, which students draw upon to develop a unique and well-justified solution to the design brief. The project is not a standalone task, but a culmination of learning that has taken place throughout the course.

As students progress, they gain experience in the design process, time management, and design for manufacture. They engage with elements of project management, concept generation, prototyping, testing, and evaluation for continuous improvement. These experiences prepare them to approach the project with both confidence and competence, enabling them to make informed decisions whilst maintaining flexibility throughout the process.

It is envisaged that the AAC will take up to 45 hours to complete. During which time, students will complete six stages of the design and manufacture process, as outlined in this document. The time allocated to each stage may vary depending on the nature of the brief and the individual student's response. Flexibility is built into the process to accommodate different approaches and solutions.

Students are required to design and manufacture a model and produce a design folio that documents their journey. The project brief, issued annually by the SEC, will include stimulus material, structural guidance, and prompts to support students in planning and completing their work. To manage workload effectively, students should begin by developing a clear work plan. Regular sharing of progress with teachers is encouraged to support authentication and provide timely feedback.

Design and manufacture are iterative processes. Students are not expected to complete the stages in a strict linear order. Instead, they may revisit activities such as brainstorming, sketching, prototyping, and testing as they refine their ideas. Mistakes and improvements are part of the process, and students should be prepared to move between stages as needed. This iterative approach reflects real-world engineering practice and supports the development of a thoughtful, well-rounded solution.

Stage 1: Analysis of Brief, Brainstorming and Developing a Work Plan

The initial stage of the Design and Manufacture Project centres on carefully reading and interpreting the design brief. Students should then conduct a critical analysis to establish a clear understanding of the context and challenges it presents. As design is fundamentally the effort to fulfil a need, identifying that need is a key element of the process. Clarifying the requirements outlined in the brief helps inform students' decisions about which project ideas to develop. It is envisaged that this stage will take approximately 3 hours to complete.

To support student learning at this stage the following activities are suggested:

- Conduct a detailed analysis of the brief to clarify project requirements and establish a foundation for further research and investigation relating to the optimum solution
- Identify key words and requirements in the brief. Students can highlight important terms and phrases, expand on their meaning, and outline their interpretation to ensure clarity
- Engage in brainstorming, individually or in groups, to generate a wide range of ideas and approaches to a design solution. Students are encouraged to explore as many ideas as possible, including ideas that initially may not appear to have potential to become a design solution. Teachers may guide the process as needed
- Record ideas using visual representations such as mind maps and spider diagrams. This helps students organise their thoughts, analyse information, and develop creative solutions. Mind maps provide structure and make complex concepts easier to understand.

Following their analysis of the design brief, students should evaluate the context and challenges it presents. This reflection helps guide their research and informs decisions about which ideas to explore further. To manage the workload effectively, students are encouraged to develop a structured work plan. This plan should break the project into manageable weekly goals, helping students organise their time and maintain steady progress. A well-developed plan supports focus, reduces pressure, and promotes student ownership of the project.

Consideration of the school calendar such as holidays, exams, and other events into the planning process helps students make realistic decisions about time allocation.

Students may find it helpful to present work plans in a graphical format, such as a timeline or Gantt chart, to clearly visualise progress.

Stage 2: Identify the focus of enquiry/research to be pursued

Through their research, students learn more about the areas that they have identified during the analysis of the context and challenges outlined in the brief. Typically, these areas include materials, components, mechanical and electronic processes, and assembly techniques. Through the use of a combination of primary and secondary research students can gain a deeper understanding of the systems and elements required to design and manufacture their optimum solution. This understanding should assist students to tailor their design to precisely match what is required from the brief, ultimately resulting in more effective and successful solutions. It is envisaged that stage 2 will take approximately 6 hours to complete.

To support student learning at this stage the following activities are suggested:

- Conduct primary research through interviews, focus groups, visits to industry, or hands-on interaction with related products or systems. Real-world insights can inspire ideas and support innovative design thinking
- Conduct further, and more focused, secondary research
- Students should reflect on their prior learning and experiences to inform their approach to the design and manufacture project. This reflection could guide their exploration of material and process sustainability, and support experimentation with different components, mechanisms, and manufacturing techniques
- Use creative ideation techniques such as SCAMPER or Think-Pair-Share to generate a wide range of possible solutions. At this stage, the focus is on idea generation, not refinement

- Begin initial prototyping and testing to gather feedback and refine ideas. Early testing provides valuable data to support design decisions. Any testing and prototyping should be carried out with due regard to health and safety.

Building on the insights gained through research, students can begin to examine their findings in greater depth, begin shaping potential solutions, while making sure all elements of the brief are fully explored.

Students can now begin to investigate the ideation of possible solutions leading to an optimum solution. This can be approached individually or as part of a group brainstorming session.

At this point in the project students need to be fully aware that during manufacturing they are provided with an opportunity to demonstrate the range and depth of skills that they have selected and developed using a range of materials, components, mechanical and electronic processes, and assembly techniques. The aim of stage 2 is to develop and present, within the design folio, a series of potential solutions that directly respond to the design brief and build upon the insights gained through research and investigation. Students may present several complete solutions to the design problem or several solutions to each of the individual problems outlined during the analysis.

A variety of techniques can be used at this stage such as:

- Students may use sketches, digital drawings, or layouts to visualise and demonstrate an idea. It is important that each idea for the solution is annotated to indicate the finer detail. This helps bring ideas to life quickly and should make it easier for others to understand and provide feedback
- Students may create prototypes or models, either digital or materials-based, to communicate the principles of operation and the physical form of their design. These prototypes can be tested and refined in terms of scale, functionality, and overall usefulness, helping to convey and develop their thinking more effectively
- Students may calculate the specifications and dimensions for the design of machine components and powered systems, if required
- Students interrogate each idea and present an evaluation of the strengths, weaknesses, critical success factors, and manufacturing methods to demonstrate how well each idea meets the context set out and challenges outlined in the brief.

Stage 3: Development of an Optimum Solution

At this stage, students begin selecting and finalising an optimum solution for manufacture. Their decisions should be guided by insights gained during research, investigation, and ideation.

Students will choose the materials and components required to produce their solution. Typically, this includes metals and polymers available in the Engineering room, though additional materials may be sourced based on their research. Students carefully evaluate which materials and processes will allow them to create a fully functioning model.

Sketches from the ideation stage should now evolve into detailed technical drawings, including dimensions and annotations that support the manufacturing process. Students must justify their choices regarding how the solution operates, how it will be manufactured, the materials selected,

and the mechanical and electronic components used. The optimum solution should also reflect an understanding of sustainable design and its environmental impact. It is envisaged that this stage will take approximately 12 hours to complete.

To support student learning at this stage the following activities are suggested:

- Develop detailed annotated sketches that illustrate all systems, subsystems, assemblies, and components required to give both form and function to the optimum solution. These sketches should clearly communicate how each part contributes to the overall design
- Create a physical or digital prototype using materials or CAD software to represent the design's form. This model should help guide decisions around scale, fit, and assembly, and support further refinement of the solution
- Describe the control system design using appropriate technical formats. This may include circuit diagrams, component modelling, or simulations using suitable software. Students can also use state machine diagrams to represent system inputs, processes, outputs, and their relationships, where relevant
- Produce dimensioned working drawings for all individual parts and assemblies. These should follow recognised drafting standards to ensure clarity and accuracy and provide all necessary information for efficient manufacture
- Compile a comprehensive parts list that includes all manufactured parts and additional components used in the physical model. The list should detail part numbers, names or descriptions, materials, key dimensions, and quantities required for assembly
- Select a simple and effective design that meets the project requirements without unnecessary complexity
- Develop a clear work plan outlining the key steps involved in the manufacturing process
- It is essential that students dedicate sufficient time and focus to fully developing their final solution before beginning the manufacturing stage. As a general benchmark, students should ask themselves: *Does my work plan, technical sketches, and dimensioned drawings contain enough detail for someone else to accurately manufacture and assemble the proposed design without further input?*

Stage 4: Manufacture of Physical Model of Optimum Solution

At this stage, students begin the physical manufacture of their optimum solution. This is a key opportunity to apply the learning developed throughout the course, demonstrating a range and depth of manufacturing skills and techniques using a variety of materials, components, processes, and assembly techniques. Health and safety must be prioritised throughout all activities. It is envisaged that this stage will take approximately 16 hours to complete.

Students are encouraged to incorporate both metals and polymers along with other suitable materials into their manufactured solution, with the form and shape determined by their design decisions. By this stage, students will have completed all necessary planning, including technical drawings, work plans, and parts lists, to confidently begin manufacture in the engineering room. It is important that students have fully developed their design before starting stage 4.

Manufacturing is an iterative process. Students may encounter design flaws or challenges that require problem-solving and the development of new ideas. It is normal to revisit earlier stages to refine aspects of the design. Mistakes should be viewed as a valuable part of the learning process. As students manufacture components, assemblies, and control systems, they should refer closely to their working drawings and parts list to ensure accuracy and consistency. The goal is to produce a fully functioning model that reflects thoughtful design and skilled execution.

In the manufacture of their optimum solution, students are expected to:

- Create a physical model to demonstrate the range and depth of manufacturing skills and techniques that they have selected to be integrated as part of the manufacturing process for their optimum solution. This may include a variety of engineering bench skills such as marking out, measuring, cutting with precision, filing, drilling, machining, and fitting. These skills should be applied across a range of materials, allowing students to showcase their understanding of material properties, process selection, and the practical execution of their design. The model should reflect not only technical ability but also the student's capacity to make informed decisions and solve problems as part of a meaningful learning experience
- Demonstrate proficiency in additive and/or subtractive manufacturing processes available in the engineering room. This may involve machining (manual or CNC), laser cutting, or 3D printing. These technologies are to be used with care and consideration for health and safety, as they can significantly enhance the quality of the final model. CAD production drawings for parts manufactured using CAM technologies are included in the design folio to support clarity and precision in the manufacturing process
- Integrate a control system or where needed, control systems into the optimum solution to showcase understanding of mechatronics and control principles from Strand 2 of the specification. The design folio should include relevant state diagrams, final circuit diagrams, and flowcharts for microprocessor-based control
- Show awareness of accuracy and finish throughout the manufacturing process. High-quality finishes can improve movement efficiency, assembly precision, and the overall appearance of the solution. Students should be familiar with appropriate finishing techniques and apply them to all faces and edges of manufactured components
- Document all manufacturing processes and the sequence followed to complete the physical model. This record should be included in the design folio and supported with photographs where appropriate
- Test the optimum solution for functionality and make modifications if required. Provide details in the design folio of all modifications made during manufacture and present reasons for such changes.

Stage 5: Testing, Evaluating and Refining

Throughout the design and manufacturing process, students engage in an ongoing cycle of exploration, revision, and improvement. Their goal is to develop a functional and well-considered physical model that responds effectively to the context and challenges outlined in the design brief.

The purpose of testing and evaluation is to confirm that the final model performs as intended and meets both functional and aesthetic expectations. If issues arise during testing, whether due to

design decisions or manufacturing errors, students may observe unexpected behaviours in the model. The evaluation should go beyond surface-level observations. It involves critical reflection, the development of insight, and thoughtful conclusions aimed at improving the process or enhancing the optimum solution.

While informal testing and adjustments may have occurred during earlier stages of manufacture, this phase requires a broader and more structured analysis. As students approach completion, they should revisit the original design brief and assess how well their optimum solution aligns with the requirements of the brief.

Meaningful learning happens when students take time to reflect deeply on their design and manufacturing journey. They should critically analyse what worked, what could be improved, and what refinements might lead to better quality or efficiency. These reflections, insights, and conclusions should be clearly documented in the design folio, supported by relevant evidence such as photographs or annotated notes.

This stage is expected to take approximately 4 hours and should be seen as an essential part of the learning process, encouraging students to evaluate their work holistically and consider how their design decisions impact performance, usability, and sustainability.

Sustainable design should be promoted to make students aware of what happens to their design or product at end of life or when decommissioned. The student should reflect on what components could be reused/upcycled/recycled to minimise wasting scarce materials and reduce environmental impact.

Stage 6: Finalising Physical Model and Design Folio

The physical model and design folio should be developed in parallel, as together they represent the student's learning across all stages of the design and manufacture project. The physical model demonstrates the range and depth of manufacturing skills and techniques the student has developed using appropriate materials, components, processes and assembly techniques, with due regard to health and safety. The design folio documents the journey from initial analysis and planning, through research, ideation, decision-making, manufacturing, testing, evaluation, and any design revisions made during production. It is envisaged that this stage will take approximately 4 hours to complete.

Students must ensure that both the model and folio are presented clearly and coherently, following the structure and section headings outlined in the project brief issued by the SEC. They should also ensure that physical model and design folio are properly presented to showcase their best work.

Particular attention should be paid to the presentation of the design folio. It should include high-quality annotated and rendered sketches, accurately dimensioned working drawings, and a complete parts list. Where applicable, students must also include state machine diagrams, circuit diagrams, and CAD drawings for any components manufactured using CAM technologies. Proof-of-concept prototypes or models, either digital or materials-based should be submitted alongside

the folio. In the physical model, students should focus on the finish and presentation of both the complete assembly and individual components.

This is the final stage in a multi-stage process that supports teacher confidence in authenticating the student's work. Teachers should not edit drafts of the folio or model. The model and folio that the student designs, manufactures, and will submit to the SEC for marking must be unique and be their own individual work, carried out in school under the supervision of the teacher. This includes both the intellectual activity of design and the practical execution of manufacture and documentation.

Teachers should also ensure that students only use machinery listed in the Department of Education and Youth Engineering Room Equipment List during manufacture. As previously advised, students should share their design folios with the teacher throughout the process to support regular check-ins and facilitate ongoing authentication.

Format and Submission of the Design and Manufacture Project

Students will be required to design and manufacture a model and produce a design folio to document the design and manufacturing process. The model and design folio will be retained in the school and presented for marking in accordance with the instructions prescribed by the SEC.

All work completed as part of the Design and Manufacture Project must be concluded by a specified date. This date will be set by the SEC and communicated to schools and will also be included in the brief that issues in Term 1 of Year 2 of the course.

The Design and Manufacture Project must be completed and submitted in line with instructions provided by the SEC. These instructions will contain specific information, such as the required structure of the design folio, section headings, page format, maximum page count, etc.

Students are required to engage in referencing to acknowledge the use of any work included in their project that is not their own. Details on referencing and using references can be found in appendix 1.

It is extremely important that all of the work submitted by students for assessment is their own. The submission of work by any student not entirely completed by that student is a significant breach of regulations. Submitting such work may lead to the imposition of penalties, up to and including the withholding of related results.

A robust authentication process is central to ensuring the integrity of any assessment process. School-based authentication by teachers of students' work on their Design and Manufacture Project is essential to the fair and equitable assessment of that work. While it is neither practicable nor necessary for teachers to witness all aspects of students' work, teachers need to be satisfied that students have carried out the work themselves. Regular, comprehensive engagement with each student's work on their Design and Manufacture Project will enable teachers to confidently and legitimately authenticate any work being submitted for assessment.

The SEC provides detailed guidance on the authentication of coursework and the conditions for its acceptance. Information as to how this applies in the case of the Design and Manufacture Project will be provided in the brief, and also in other documentation which may be issued by the SEC.

Descriptors of Quality

The descriptors below relate to the learning achieved by students in the design and manufacture project. In particular, the design and manufacture project require students to:

- Analyse, Research and Plan
- Develop Ideas
- Manufacture Solutions
- Test and Evaluate
- Document the Learning Journey.

Table 2: Descriptors of quality: Design and Manufacture Project

	Students demonstrating a high level of achievement	Students demonstrating a moderate level of achievement	Students demonstrating a low level of achievement
Analysis, Research and Planning	Conduct an in-depth exploration of the research, analysis, and planning stages, demonstrating a sophisticated understanding of the brief. Key specifications and constraints are thoroughly examined.	Engage thoroughly in the research, analysis, and planning stages, showing a clear and practical understanding of the brief. Consideration is given to most key specifications and constraints.	Participate in the research, analysis, and planning stages, showing a basic and general understanding of the brief. Some key specifications and constraints are acknowledged.
Developing ideas	Present a solution that thoroughly communicates their ideas and concepts, addressing all aspects of the brief comprehensively. Develop a highly effective and detailed workable solution.	Offer a solution that clearly conveys their ideas and concepts, covering most aspects of the brief adequately. Create a functional solution with sufficient detail.	Offer a solution that communicates their ideas and concepts to a basic extent, addressing some aspects of the brief. Produce a workable solution with minimal detail.

Manufacturing Solutions	Manufacture a solution that exemplifies a broad and proficiency of technical skills and processes, achieving high levels of functionality, accuracy, precision, and a superior finish.	Manufacture a solution that reflects a command of technical skills and processes, ensuring good functionality, accuracy, precision, and a quality finish.	Manufacture a solution that demonstrates fundamental technical skills and processes, with limited functionality, accuracy, precision, and a satisfactory finish.
Testing and Evaluation	Test and evaluate their solution showing a detailed and critical reflection of their design and manufacturing journey with clear recommendations for potential improvement.	Test and evaluate their solution showing moderate detail and some reflection of their design and manufacturing journey with some recommendations for potential improvement.	Test and evaluate their solution showing limited detailed and limited reflection of their design and manufacturing journey with limited recommendations for potential improvement.
Documenting the Learning	Present a folio presentation that thoroughly communicates and justifies their design and manufacturing decisions, demonstrating a high level of detail and insight.	Present a folio presentation that effectively communicates and justifies their design and manufacturing decisions, showing a moderate level of detail and understanding.	Present a folio that communicates their design and manufacturing decisions with limited detail and justification.

Role of the teacher

The teacher has an important role in facilitating, supporting and supervising student work on the Design and Manufacture Project. Teaching and learning related to the additional assessment component should be integrated into ongoing classroom practice to maximise opportunities for students to achieve the learning outcomes of the specification and support the development of key competencies.

As stated previously, students will be acquiring the skills which prepare them for the completion of the Design and Manufacture Project from the very beginning of the course, as they engage with the learning outlined in the specification, and develop the knowledge and skills required to complete their unique optimum solution to the design brief. The release of the common brief by the SEC in term one of year two allows teachers to provide guidance and clarification regarding the requirements and content of the brief whilst facilitating teachers in planning for the

integration of the Design and Manufacture Project, as appropriate, into the natural flow of teaching and learning in the classroom.

Support may include:

- clarifying the requirements of the Design and Manufacture Project brief
- helping students to understand how the brief issued by the SEC for the Design and Manufacture Project links to the Engineering specification
- identifying any particular areas of the specification that students may need to engage with in order to undertake the Design and Manufacture Project brief
- ensuring students are aware of the descriptors of quality associated with the Design and Manufacture Project brief
- offering prompt questions, such as those in Appendix 2, to stimulate and support students' planning and critical thinking throughout the process
- encouraging students to focus on the requirements of the brief and to relate their work to the learning outcomes in the specification
- guiding students in the continued development of the skills associated with design and manufacture
- helping students to appreciate the importance of good referencing practices and the necessity to avoid any form of plagiarism
- facilitating access to appropriate resources
- providing appropriate access arrangements for students with additional learning needs, in accordance with the SEC's provision of reasonable accommodations
- making students aware that they must comply with any relevant mandatory regulations or requirements such as those relating to health and safety, General Data Protection Regulations (GDPR), accessing public or private property etc.
- using feedback appropriately to promote a reflective and iterative approach to work on the Design and Manufacture Project
- providing instructions at strategic intervals to facilitate the timely completion of the Design and Manufacture Project
- drawing students' attention to any requirements in relation to the submission of the Design and Manufacture Project specified in the brief
- promoting opportunities afforded by the Design and Manufacture Project to develop Senior Cycle key competencies.

In order to facilitate the authentication process, teachers should engage regularly with students' work on the Design and Manufacture Project. Teachers should be aware that only work which is the student's own can be submitted for assessment to the State Examinations Commission and that each student must submit their work on an individual basis. The feedback provided to students should be general and nondirective in nature. Teachers should not provide any excessive or inappropriate support to students, such as editing draft work or providing model text or answers to be used in the students' design folio.

Appendix 1: Guidelines to support referencing

Referencing is an important aspect of the AAC as it allows those reading the Design Folio to better engage with the content and to verify the information provided in the reference. It is the most appropriate way for students to acknowledge the source of any information, ideas, material or images not their own which they have included in their Design Folio.

Referencing allows students to provide evidence of the research they have engaged in, it helps to support and give weight to arguments and conclusions, and it can be used to demonstrate that different perspectives have been considered and explored by the student.

Students should engage in referencing both within the body of their work (in-text citation) and also in the reference section of the Design Folio. The use of in-text citation provides a direct link between what students write and the research on which their work is based.

Example: A student using material from page 57 of the book *Inclusion: effective practice for all students?* would use the in-text citation (p. 57, McLeskey, 2013) and then go on to also add details to the list of references as McLeskey, J. (2013) *Inclusion: effective practice for all students?* 2nd edn.

In the reference section of their document, students should provide the appropriate details of any sources they have used during the course of their Design and Manufacture Project such as:

- books, newspapers, magazines
- professional journals and government reports
- online sources including videos, podcasts etc.
- material from specialist organisations and relevant individuals
- material generated by artificial intelligence (AI) software and AI applications.

Specific information will be issued around this in the Design and Manufacture Project brief and in related documentation from the SEC.

The reference section is not included in the word count. Comprehensive referencing helps to show that students have engaged in honest and ethical research practices and have avoided plagiarism. Referencing should be as specific as necessary to communicate the particular research source, such as a page number or chapter in a book, a section in a website, timestamp on a podcast or video etc.

Plagiarism is a serious offence and occurs when work other than the student's own is used without clear acknowledgement of the source of the work. This includes the use of material generated using artificial intelligence (AI) software or AI applications. Direct copying of material from any source without proper acknowledgement is not permitted and may incur penalties, up to and including the withholding of related results.

When referencing the sources students used in their report they should ensure that, regardless of the type of source, there is enough accurate detail to enable the reader to authenticate the reference. No particular, formal style of referencing is required.

How students should reference different sources

- Where students are citing written sources or information in print, they should give the author's name, the title of the publication, year of publication, and, if necessary, the page number or chapter/section of the publication

- Where students wish to refer to an internet site or online source, there should be enough accurate detail to enable the reader to authenticate the reference, including the hyperlink and date read or downloaded
- Where students have used material generated by artificial intelligence (AI) software and AI applications this must be acknowledged. The reference should include the name of the AI tool used, the date the content was generated and provide a brief explanation of how it was used. Many Generative AI tools generate shareable URLs that set out the content of chat sessions that took place. Where such a tool has been used, the URL should be included in the list of research sources. Where an AI tool does not generate a sharable URL, student should include the name of the tool and the prompt used.

Examples of in-text Citations for different types of sources
Book: (p. 57, McLeskey, 2013)
Newspaper/magazine article: (Hearne, J., 30/08/2024)
Text/image accessed online: (thelatinlibrary.com/101/RhetoricalDevices)
Audio accessed online: (Ep. 10, rte.ie/radio/podcasts/22093250)
Video accessed online: (3:20 to 5:45,youtu.be/yCv4iyPqZKQ)
AI Tools (with shareable url): (chat.openai.com/share/f45a1e23-2217-4443-a244-d56ab26ae940)
AI Tools (without shareable url): (OpenAI (2023) ChatGPT, 20/10/2023)

Appendix 2: Suggested prompt questions to support students in the process for completion of the Design and Manufacture Project

Stage 1: Analysis of Brief, Brainstorming and Developing a Work Plan

During initial analysis, brainstorming, and developing a work plan, students may consider:

- Have I clearly identified the specifications prescribed by the brief? The key features and/or functions the model should have?
- What exactly must my model be able to do?
- Are there other ways these specifications could be interpreted?
- What problem is trying to be solved?
- Have I seen this problem being solved before? Where? How was it solved? What can I learn from this?
- What constraints does the brief outline?
- How much time is available? Are there any upcoming events that might interfere with the available time?
- How much time should I allocate for each task?

Stage 2: Identify the focus of enquiry/research to be pursued

When undertaking the research, investigation, and ideation of possible solutions, it may be helpful to consider:

- What exactly must my model be able to do?
- Who might use this solution?
- What problem is it solving?
- What ideas worked well for others?
- What possible primary sources of information I could find?
- What possible secondary sources can I use to gain more understanding?
- What other ways could the same motion/movement/effect be achieved?
- What mechanical or electrical components will I need to make it work effectively?
- How could I categorise this concept/idea:
 - It should work
 - It might work
 - It won't work
 - I don't know
- What similar products, systems, or technologies already exist?
- What materials are suitable to use, based on their properties?
- Are suitable materials available in the engineering room, or can they be sourced?
- How does the chosen materials affect the choice of manufacturing processes required, and the end of life recyclability?
- Have I referenced and acknowledged all research sources?

Stage 3: Development of an Optimum Solution

As the student compiles and refines their initial designs, bringing the most effective ideas for individual parts into one coherent, unique design for an optimum solution, they may consider the following:

- Have I addressed all the requirements of the brief?
- Is the basic engineering/concept sound?
- Do manufacturing approaches exist?
- Does my work plan allocate time for the necessary steps in the manufacturing process that I can think of?
- Can it be manufactured/completed in the allocated time constraints?
- How could I best organise my workflow for manufacture; what needs to happen first, the time constraint needed; next step, etc in a logical sequence, mindful of Parkinson's law – the amount of work will expand to fill the time available.
- Is the concept consistent with the rest of the design?
- Could the design be refined to ease/assist assembly?
- What mechanical or electrical components will I need to make it work effectively, and are there associated calculations to support this required?
- How will the operator interact with, or control this model? Simplicity and intuitiveness of use, ergonomics, use of visual, audio, or haptic feedback, etc.
- Have I created detailed annotated sketches of all systems, subsystems, assemblies, and components necessary to give form and function to their optimum solution.
- Can I create a material or CAD prototype/model to communicate the physical form of my design?
- Is it possible to model my control system and circuit with components, or simulation in suitable software?
- Can I create dimensioned working drawings of all individual parts and assemblies?
- Have I created a parts list/bill of materials for my model/prototype?
- Is my design as simple and effective as I can make it, being mindful of over-engineering a model?
- Is all the necessary detail presented (in technical sketches, dimensioned working drawings, etc.) so that I can now fully manufacture and assemble my proposed design solution?
- Will my optimum solution demonstrate the range and depth of skills that I have developed in Engineering?

Stage 4: Manufacture of Physical Model of Optimum Solution

As the student compiles and refines their initial designs, bringing the most effective ideas for individual parts into one coherent, unique design for an optimum solution, they may consider the following:

- Does my physical model demonstrate a range and depth of manufacturing skills and techniques using an assortment of materials, components, mechanical and electronic processes, and assembly techniques?
- Does my physical model include a selection of the bench skills, such as marking out and measuring, precision filing, drilling, machining and fitting?
- Is my model manufactured using a range of appropriate materials?
- Am I demonstrating proficiency in a range of additive and/or subtractive manufacturing processes readily available in the engineering room?

- Do all manufactured components have high levels of accuracy and finish, on both faces and edges?
- Have I kept detailed record of all manufacturing processes and the sequence used to complete the physical model (include photos where appropriate)?
- Have I kept a record of any tests made for functionality and reasons for refining design decisions?

Stage 5: Testing, Evaluating and Refining

During testing and evaluation, students may consider:

- How did the design process go?
- How have I benefitted from this process, in terms of knowledge, skills, values, and dispositions?
- Did the workplan help me manage the project workload efficiently?
- Does the model meet the criteria of the design brief?
- Does it meet the engineering specifications that I set for myself?
- Does the model look good? Scale well?
- Does it look/act as I intended?
- What errors were in the design?
- What errors were in the manufacturing, i.e. beyond my control?
- How could I have improved quality or efficiency?
- Are the components up to spec, e.g. motors too fast/slow/strong enough?
- Is the Human Machine Interface (HMI) easy to use? intuitive? ergonomic? safe? Tidy? Attractive? etc.

Stage 6: Finalising Physical Model and Folio

When finalising the physical model and folio, students may consider:

- Are the physical model and design folio presented in accordance with the required structure, and in a coherent manner?
- Are the section headings and other details outlined in the project brief addressed?
- Have I included high quality annotated and rendered sketches?
- Have I produced accurately dimensioned working drawings? (including CAD drawings if CAM is used to manufacture a specific component in the model)
- Have I included a complete parts list?
- Have I included state machine (if necessary) and circuit diagrams?
- Have I cardboard or other proof-of-concept prototypes presented for assessment?

Appendix 3: Prototypes

The following appendix is adapted from Ullman (2010), *The Mechanical Design Process*¹, to support student understanding of prototyping in engineering.

Prototypes

Physical models of products are often called prototypes. In the engineering design process, the terms modelling and prototyping are often interchangeable, however, the function of a prototype is universally recognised for testing/evaluation and so prototyping is an essential activity for an engineer. The student will ultimately produce a physical model, which is a fully functioning prototype, of their design for assessment. They should be encouraged to make other prototypes while developing and refining their design as they gain new ideas and further understanding of concepts through research and investigation. Prototypes can range from low accuracy cardboard/paper models, up to high precision functional machines and it is standard practice in engineering fields such as automobile and aircraft manufacture. They provide the student with valuable learning experience in manufacture and an effective method of communicating their design ideas. Prototyping traditionally applied only to physical models; however, CAD models and simulation software can replace some physical prototypes with less cost and time. Prototypes are categorised by their intended purpose, the stage of the design process when they are used, and the media used to build them. The main categories that could be used are:

A **proof-of-concept** or **proof-of-function** prototype which focuses on developing the function of the product for comparison with the criteria of the brief, or engineering specifications. This kind of prototype is intended as a learning tool, and exact geometry, materials, and manufacturing process are usually not important. Proof-of-concept prototypes are generally low quality, low cost, and fast to make. They can be made from paper, wood, recycled parts from toys, or whatever is handy. They are useful for students to gain a sense of form and scale for their designs, as well as testing mechanisms/linkages etc. for the required function.

A **proof-of-product** prototype is developed to help refine the components and assemblies. Geometry, materials, and manufacturing process are as important as function for these prototypes. The recent development of rapid prototyping or desktop prototyping, using 3D printing or other methods to produce a part rapidly from CAD software, has greatly reduced the time and cost efficiency of building proof-of-product prototypes. Similarly, circuit simulation software allows a student to build and test a range of components and circuits or code for required function at minimal cost and no dangers of damaging expensive or scarce parts.

A **proof-of-process** prototype is used to verify both the geometry and the manufacturing process. For these prototypes, the exact materials and manufacturing processes are used to manufacture samples of the product for functional testing. A student may use this type of prototype to ensure that the component they have designed will assemble as intended. For example, if a part has been designed on a CAD system, the student may have it laser cut in card or heavy paper to test for an accurate fit of various components such as motors, axles, etc. on the part before laser cutting in acrylic, potentially saving time and material waste while providing practice and reassurance for the assembly process.

¹ Ullman, D. G. (2010). *The mechanical design process* (4th ed.). McGraw-Hill.

A **proof-of-production** prototype is the completed physical model submitted for assessment by the student. It is a fully functioning prototype of their optimum solution for the design challenge. This finished prototype/model is used to verify the entire production process and is complete with all the required materials, processes, mechanical and electrical components assembled and wired. This prototype should function as the student's design intended.



NCCA

An Chomhairle Náisiúnta
Curaclaim agus Measúnachta
National Council for
Curriculum and Assessment